Real case: one defect nobody sees costs

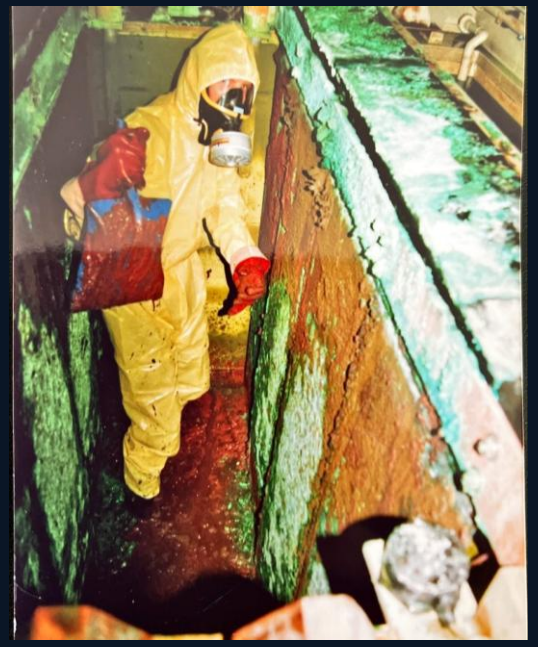
€350,000

170 t CO_2

Wouldn't it be better to act — before the damage is done?

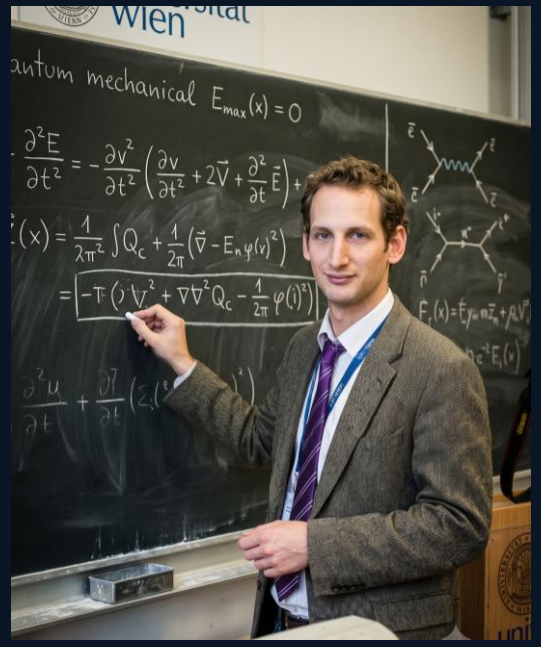
20 years of learning — to make this possible

2005



Apprentice to master
Surface finishing

2014



Physicist
MSc, University of Vienna

2020



IT process consultant
& lean manager

2025



Founder
secure galvano ai

The invisible time bomb at your customer

Shipped and assembled undetected — recall and scrap



The flood of data in manufacturing

Every year, modern production lines generate enough data to stack as high as a **church tower.**

Until now, the **technology** to analyse it was missing.



Skills shortage and firefighting

50% of the skilled workers the industry needs are missing today

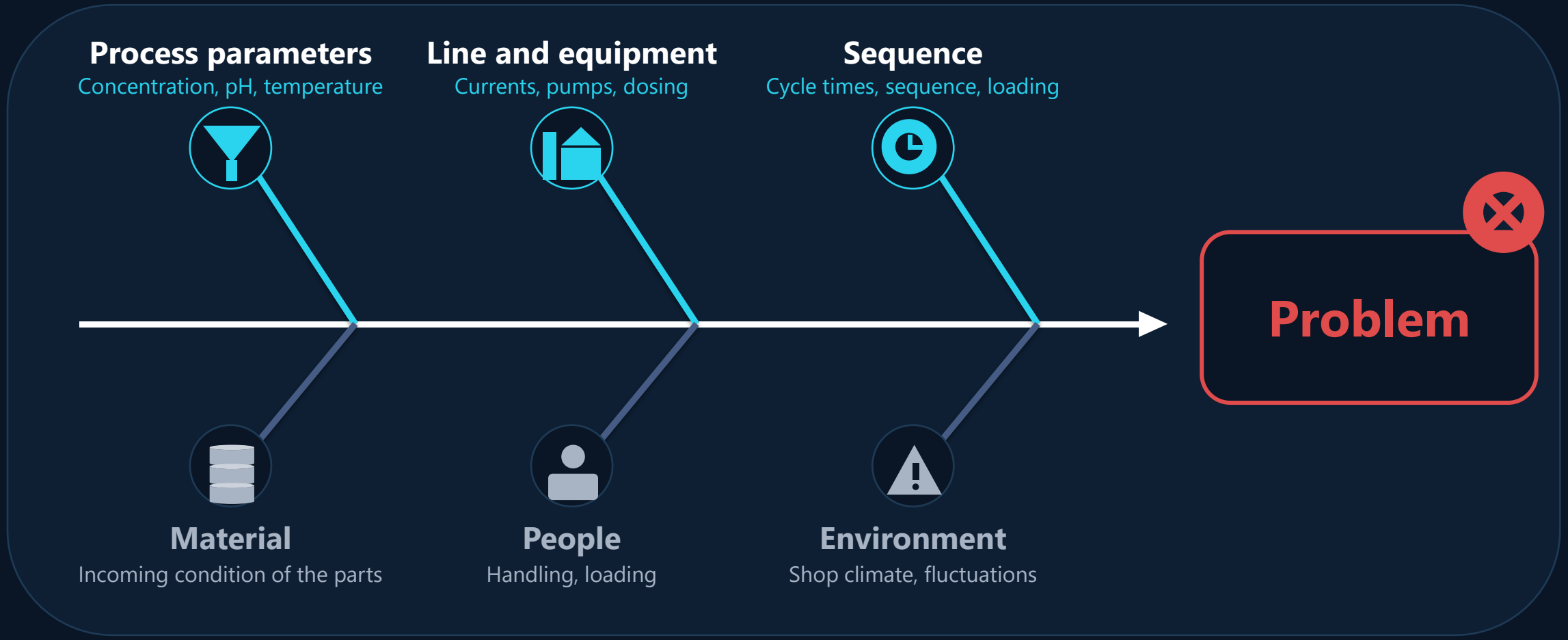
Defects are only **fixed on the surface** — and keep coming back

Root cause analysis in complex processes takes **weeks to months**



The solution: narrowing down causes automatically

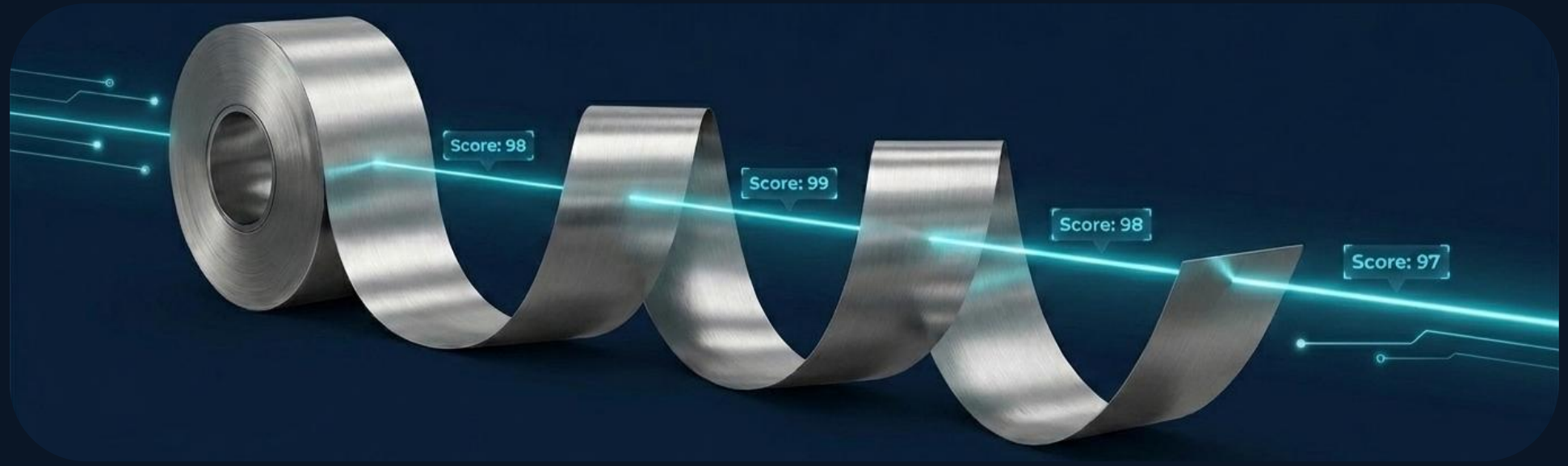
The software checks what the process data holds — your specialists settle the rest



● evidenced in process data

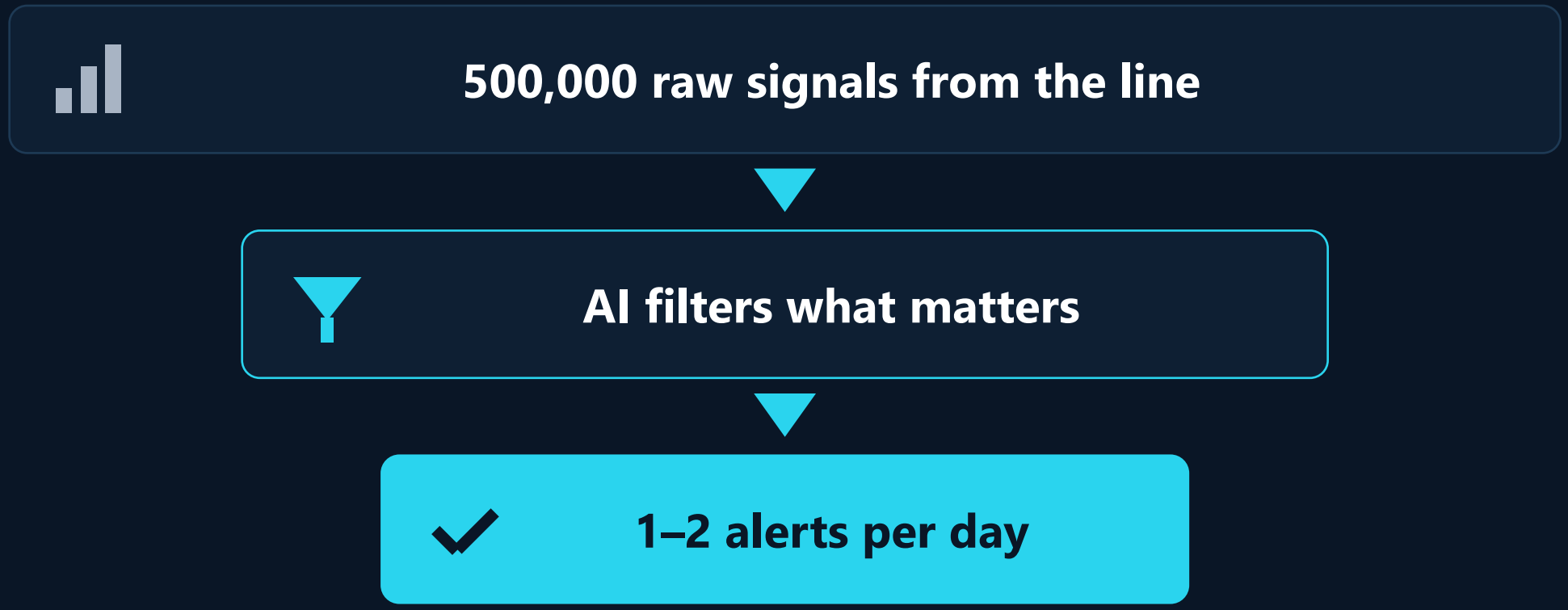
● with your specialists (experience)

Predictive Quality: one quality score per batch



From your process data we calculate the **quality condition of every batch** — so that you can act **before major damage is done.**

The flood of data becomes clear alerts



Many signals become a few clear alerts a person can act on.

Evidence from the pilot project

Here from an electroplating shop — the method fits any process with measurement data





real defect on the part

Flagged batches are inspected before they leave the building — no recalls.

Second example: a Site Assessment at another customer

The most likely causes of a defect pattern — ranked automatically

ERP: defective parts per batch

Data extract from the line controls

Local AI analysis



Merkmal	Warenträger mit Merkmal	Fehlerbild 14 % mit Merkmal	% ohne Merkmal	x-mal so häufig
Meldung 4.8 Unzulässige Regelabweichung	223	10,66	1,52	7,02
Meldung 3.11 WT-Nr. in Position Gleichrichterstörung	227	10,17	1,52	6,69
Gleichrichter Messgröße 5 höher als üblich	292	4,11	1,47	2,79
Gleichrichter Messgröße 3 höher als üblich	177	3,89	1,54	2,53
Gleichrichter Messgröße 6 höher als üblich	390	3,42	1,47	2,33
Gleichrichter Messgröße 0 niedriger als üblich	281	2,28	1,62	1,41
Gleichrichter Messgröße 0 höher als üblich	84	1,84	1,64	1,12
Temperatur Messgröße 1 niedriger als üblich	160	1,80	1,64	1,10
erster Warenträger nach Stillstand (> 30 min)	982	1,73	1,62	1,07

Main causes of a defect pattern

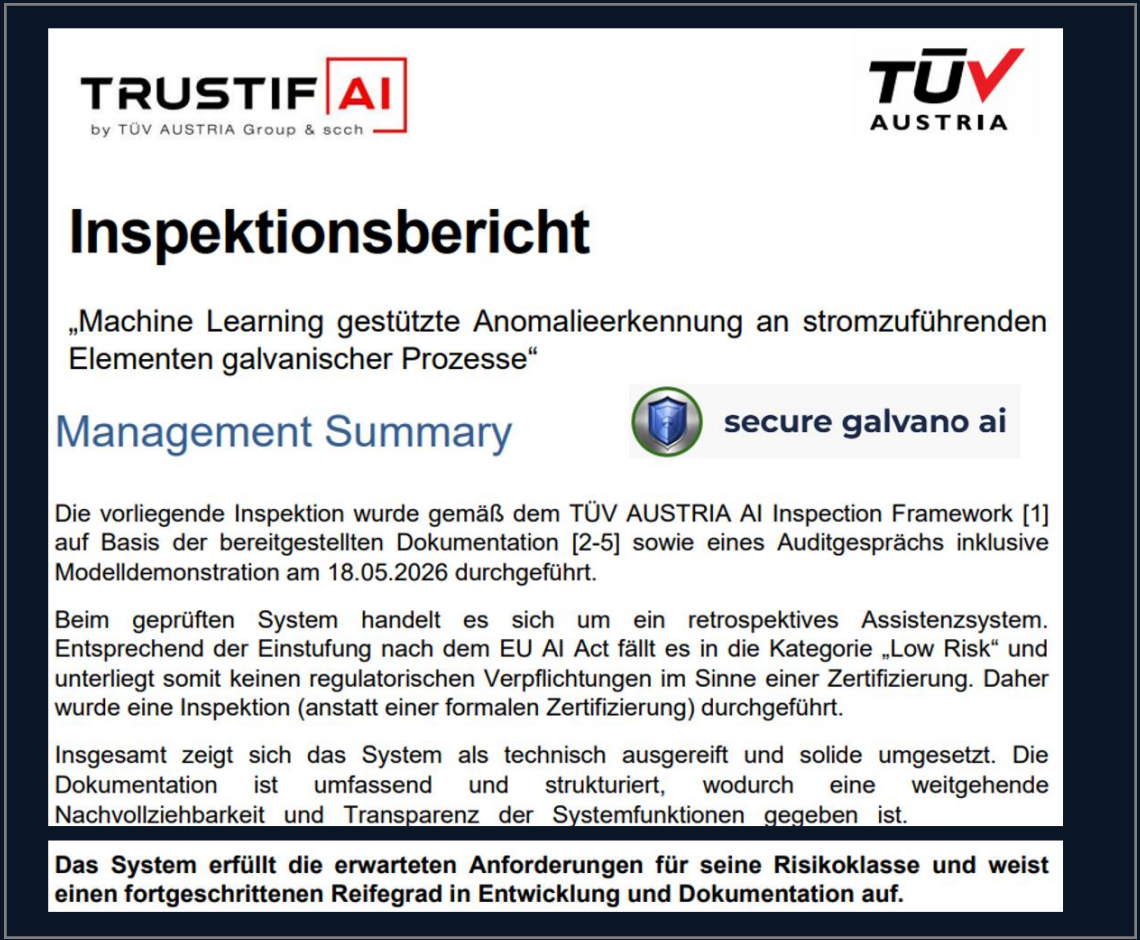
Your effort

1 h briefing · 30 min data extract

On our side

3 days to the first analysis — after that it runs automatically for each defect pattern

Trust through independent audit



TRUSTIF AI
by TÜV AUSTRIA Group & scch

TÜV AUSTRIA

Inspektionsbericht

„Machine Learning gestützte Anomalieerkennung an stromzuführenden Elementen galvanischer Prozesse“

Management Summary

secure galvano ai

Die vorliegende Inspektion wurde gemäß dem TÜV AUSTRIA AI Inspection Framework [1] auf Basis der bereitgestellten Dokumentation [2-5] sowie eines Auditgesprächs inklusive Modelldemonstration am 18.05.2026 durchgeführt.

Beim geprüften System handelt es sich um ein retrospektives Assistenzsystem. Entsprechend der Einstufung nach dem EU AI Act fällt es in die Kategorie „Low Risk“ und unterliegt somit keinen regulatorischen Verpflichtungen im Sinne einer Zertifizierung. Daher wurde eine Inspektion (anstatt einer formalen Zertifizierung) durchgeführt.

Insgesamt zeigt sich das System als technisch ausgereift und solide umgesetzt. Die Dokumentation ist umfassend und strukturiert, wodurch eine weitgehende Nachvollziehbarkeit und Transparenz der Systemfunktionen gegeben ist.

Das System erfüllt die erwarteten Anforderungen für seine Risikoklasse und weist einen fortgeschrittenen Reifegrad in Entwicklung und Dokumentation auf.

TÜV AUSTRIA / TRUSTIFAI

Model audit completed: 2 June 2026

Classification:
EU AI Act low risk

Aligned with:
Information security — ISO/IEC 27001
AI management — ISO/IEC 42001

Local data processing

Thank you to the people who made this possible

aws AI Adoption Green project · 06/2025 – 08/2026 · project volume €190k

06/2025 — Start with simulated data

12/2025 — On a real production line

03–05/2026 — Evaluated in operation

06/2026 — Model audit, TÜV AUSTRIA

CONTRIBUTORS



Brigitte Maier
Strategy & business development



Sandra Basic
Finance & HR



Fabian Kraft
Data engineering



Jürgen Paulmichl
Data security



Verena Becker
Cybersecurity, NIS



Jenny Bors, BSc
Innovation protection & IP management



Christine Knecht-Kleber
Legal



Gertraud Wahrlich
Tax advice

AUDIT AND FUNDING



Barbara Brune
Statistics & data science, TÜV AUSTRIA



Dr. Regina Weinmüllner
Funding support, aws

No one carries a project like this alone — **thank you!**

What your operation gains

Four effects — in any plant that records its process data



Less manual inspection

▼

Lower labour costs



Catch defective batches

▼

Protect customer trust



React faster

▼

Lower cost of defects



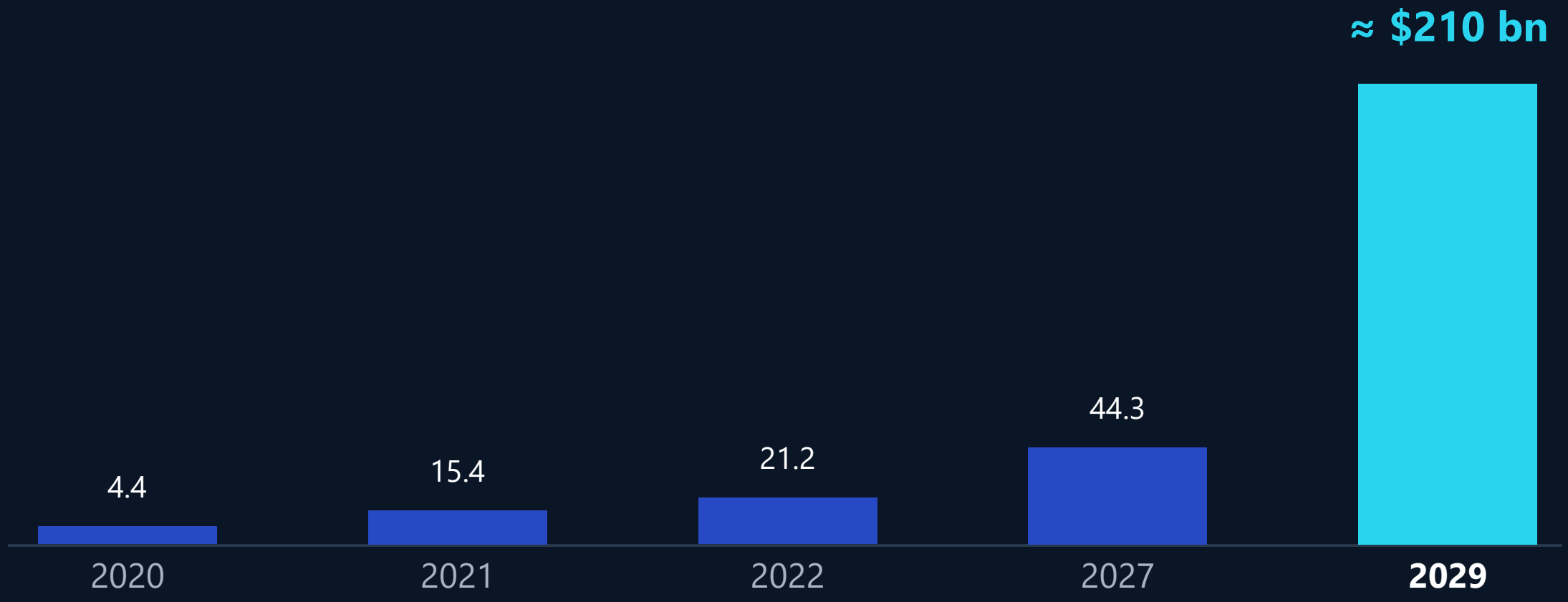
Minimise waste

▼

Reduce the CO_2 footprint

Machine learning market

Global market in bn US dollars — tenfold growth from 2022 to 2029



Source: 45+ Machine Learning Statistics — Market Size, Industry, Growth (2026)

How the collaboration grows

1

Site Assessment

ERP extract with scrap data
Data extract from the line
Your effort: 2 h

€1,500

2

Proof of Concept

One specific use case
Your effort: 5–10 h

€5,000

3

Solving the problem

3–4 months of support
through to the solution
Your effort: per project

time & materials

THE POTENTIAL PER USE CASE AND YEAR

€60–90k

Resources, labour, disposal costs, CO_2

Payback < 1 year

Your starting point

How to start — remote or on site



First find out where the gold is. Then dig.

About half a day for your people, spread over two weeks.



The intro call

free · 30 minutes

There we assess your case: whether you are still looking for where the money is, or already have a specific question.

›

Site Assessment

€1,500

Unlock the treasure in your data!

Book a free 30-minute intro call. We assess your case, with no obligation.

Contact: Stefan Maier

secure galvano ai

Web	secure-galvano-ai.com
Email	smaier@rvh.at
Phone	+43 681 8148 3538



Learned from the master. Validated by the physicist. Automated by AI.